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Presentation Outline

1. Introduction & Theoretical Framework
2. Methodology
3. Research Findings
4. Discussion & Pedagogical Implications
5. Limitations & Conclusion

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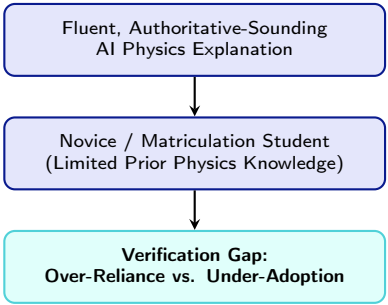
Background: The AI-EdTech Dilemma in Physics

The Adoption Lag

- AI-EdTech offers personalised feedback, gap diagnosis, and adaptive STEM support (Nazaretsky et al., 2025).
- Adoption lags behind technical capability due to **social and psychological barriers** rather than functional limits (Nazaretsky et al., 2025).

The Physics-Specific Verification Trap

- Generative AI is **non-deterministic and opaque** (Nazaretsky et al., 2025).
- Students rate AI linguistic quality nearly as high as expert solutions, but struggle to detect scientific errors when prior knowledge is limited (Dahlkemper et al., 2023).
- 68.2% of physics students express concern over inaccurate AI outputs (Veith et al., 2026).



Theoretical Framework: Extended TAM & Ambivalence

1. Extended TAM: Trust as Prerequisite

- **Classical TAM**: Davis (1989) assumes deterministic, knowable behaviour (Usefulness + Ease of Use).
- **Trust**: Lee and See (2004) define trust as the attitude that an agent will help achieve goals under *uncertainty* and *vulnerability*.
- **Extended TAM**: In opaque AI systems, Perceived Trust directly predicts Perceived Usefulness, $\beta = .45$ (Nazaretsky et al., 2025).

2. Psychology of Ambivalence

- **Unidimensional Flaw**: Collapsing positive and negative items hides simultaneous mixed feelings (Schneider & Schwarz, 2017).
- **Bidimensional Approach**: Schepman and Rodway (2026) showed subscales correlate only moderately ($r = .485$) with distinct predictors.
- Captures **ambivalence** (high positive + moderate negative) vs. neutrality.

Problem Statement & Spatial Gap

Global physics studies converge on **trust** over raw usefulness (Agyare et al., 2025; Wattanakasiwich et al., 2025), yet **no psychometrically validated instrument** measures AI trust alongside bidimensional attitudes in a **Malaysian pre-university matriculation** population.

Research Objectives & Questions

Focus	Research Objectives (RO)	Research Questions (RQ)
1. Psychometrics	**RO1:** Evaluate construct validity (EFA) and internal consistency (α, ω) of the physics-adapted Short GAAIS-10 and AI-EdTech Trust Scale.	**RQ1:** Do the adapted instruments demonstrate acceptable construct validity and reliability among Malaysian physics matriculation students?
2. Demographics	**RO2:** Identify the demographic and AI-usage profile of the sample.	**RQ2:** What baseline attitudes (positive and negative) do students hold toward AI in physics learning, and do they vary by AI usage frequency or gender?
3. Attitudes	**RO3:** Determine baseline positive and negative attitudes toward AI in physics learning across usage frequency and gender.	
4. Trust	**RO4:** Determine baseline level of trust in AI-EdTech for physics learning across usage frequency and gender.	**RQ3:** What level of trust do students place in AI-EdTech for physics learning, and does it vary by AI usage frequency or gender?

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Methodology: Research Design & Sample Profile (RO2)

Study Design & Sampling

- **Design:** Quantitative, cross-sectional survey.
- **Site:** Sarawak Matriculation College, Ministry of Education Malaysia.
- **Sample:** $N = 88$ students (Semester 2, 2025/2026 cohort; convenience sampling).
- **Ethics:** Institutional approval & informed consent; zero missing data.

Key Takeaway (RO2)

70.4% of respondents use AI *"Often"* or *"Very Often"*, indicating high baseline exposure.

Table 1 *Demographic Profile of Respondents*

Variable	Category	n	%
Gender	Female	51	58.0
	Male	37	42.0
Module	Sains Hayat (Life Science)	60	68.2
	Sains Fizikal (Physical Science)	28	31.8
AI Usage	Often	50	56.8
	Sometimes	23	26.1
	Very Often	12	13.6
	Rarely	3	3.4

Note. $N = 88$ **for all variables; no missing data were reported.**

Physics-Contextualised Instrument Adaptation

Table 2*Illustrative Item Adaptations From Source Instruments to the Physics-Learning Context*

Source	Original Wording	Adapted	Physics-Adapted Wording
Pos7	I am interested in using artificially intelligent systems in my *daily life*.	**B1**	I am interested in using artificially intelligent systems in my **daily physics learning**.
Pos18	I would like to use artificial intelligence in my *own job*.	**B5**	I would like to use artificial intelligence in my **own physics learning**.
Neg8	I find artificial intelligence sinister.	**B6**	I find artificial intelligence sinister. (unchanged)
PB1	AI can assist me in improving my responses (e.g., open-ended questions, *writing essays*)...	**C1**	AI can assist me in improving my responses to **physics problems** (e.g., open-ended questions, **calculations**)...
PB6	AI can assist me in preparing for exams by identifying my knowledge gaps...	**C6**	AI can assist me in preparing for **physics exams** by identifying my knowledge gaps...
PT4	The more autonomy AI-based tools provide *me/my teacher* to check/overwrite...	**C10**	The more autonomy AI-based tools provide **me** to check or overwrite...

***Note*. Source items for Section B (Pos7, Pos18, Neg8) are from Schepman and Rodway (2026); Section C (PB1, PB6, PT4) from Nazaretsky et al. (2025). B6 and C10 reflect minor adaptation deviations; both loaded cleanly (.631 and .747).**

Data Analysis Procedure (in Jamovi)

1. Psychometric Evaluation (RO1)

- **Exploratory Factor Analysis (EFA):**
  - Principal Axis Factoring (PAF) extraction.
  - Oblimin (oblique) rotation (Schepman & Rodway, 2026; Nazaretsky et al., 2025).
  - Threshold: Cross-loadings $< .30$.
- **Sampling Adequacy:** Kaiser-Meyer-Olkin (KMO) and Bartlett's test of sphericity.
- **Internal Consistency:** Cronbach's α and McDonald's ω with bootstrapped item analysis.

2. Baseline & Group Comparisons (RO3–RO4)

- **Normality Check:** Shapiro-Wilk tests showed significant departures from normality ($p \leq .027$).
- **Group Comparisons:** Non-parametric **Kruskal-Wallis** tests (χ^2, effect size ε^2).
- **Post-Hoc Tests:** Dwass-Steel-Critchlow-Fligner (DSCF) pairwise comparisons.
- **Bivariate Associations:** Spearman's rank-order correlations (ρ).

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Construct Validity: Exploratory Factor Analysis (RO1)

Table 3 *EFA Loadings for the Attitudes Scale (B1–B10)*

Item	F1 (Positive)	F2 (Negative)	Uniqueness
B1	**.832**	—	.309
B2	**.742**	—	.450
B3	**.823**	—	.317
B4	**.808**	—	.347
B5	**.826**	—	.283
B6	—	**.631**	.518
B7	—	**.712**	.479
B8	—	**.764**	.384
B9	—	**.779**	.381
B10	—	**.635**	.591

Note. **PAF extraction, oblimin rotation. KMO** $= .804$; **Bartlett's** $\chi^2(45) = 468, p < .001$. **Cumulative variance** $= 59.4\%$ **(F1**: 34.1%, **F2**: 25.3%**)**.

Table 4 *EFA Loadings for the Trust Scale (C1–C10)*

Item	F1 (Usefulness)	F2 (Trust/Intent)	Uniqueness
C1	**.770**	—	.412
C2	**.808**	—	.368
C3	**.752**	—	.401
C4	**.833**	—	.366
C5	**.665**	—	.468
C6	**.758**	—	.384
C7	—	**.502**	.386
C8	—	**.598**	.299
C9	—	**.987**	.137
C10	—	**.747**	.350

Note. **PAF extraction, oblimin rotation. KMO** $= .858$; **Bartlett's** $\chi^2(45) = 605, p < .001$. **Cumulative variance** $= 64.3\%$ **(F1**: 40.3%, **F2**: 24.0%**)**.

Internal Consistency & Baseline Levels (RO1, RO3, RO4)

Table 5 *Reliability and Descriptive Statistics for Study Constructs*

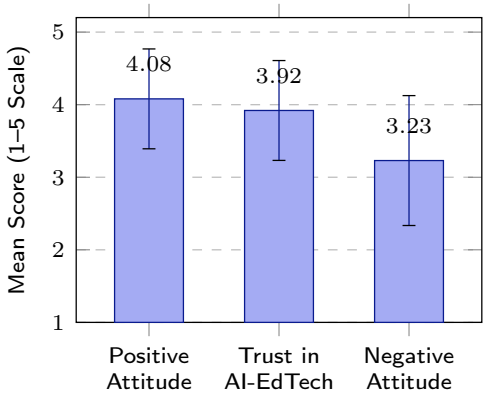
Construct (Items)	α	ω	M (SD)	Mdn	Level
Pos. Attitude (B1–B5)	.902	.903	4.08 (0.69)	4.00	**High**
Neg. Attitude (B6–B10)	.825	.828	3.23 (0.90)	3.20	**Mod.**
Trust in AI (C1–C10)	.921	.922	3.92 (0.69)	3.95	**High**

Note. $N = 88$. **Means interpreted relative to scale midpoint (3.00).**

Considered Optimism (Ambivalence)

- High Positive Attitude ($M = 4.08$) and Trust ($M = 3.92$) coexist with moderate Negative Attitude ($M = 3.23$).
- Exceeds prior university trust baselines ($M = 3.31$ in Nazaretsky et al., 2025; $M = 3.47$ in Dahlkemper et al., 2023).

Figure 1 *Mean Scores for Study Constructs*



Note. **Error bars represent standard deviations (**$N = 88$**).**

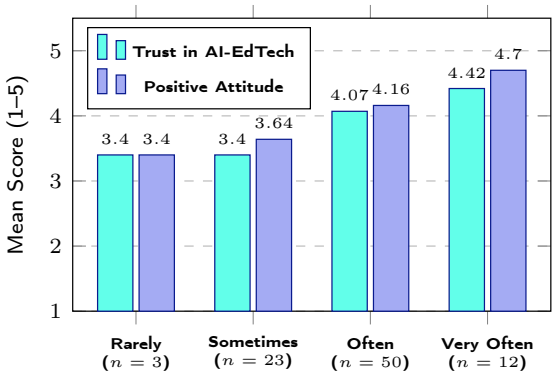
Group Differences: AI Usage Frequency & Gender (RO3, RO4)

Table 6 *Kruskal-Wallis Test Results by Usage and Gender*

Construct	Factor	χ^2 (*df*)	p	ε^2
Trust	AI Usage	22.53 (3)	$< .001$	**.259**
Pos. Attitude	AI Usage	22.44 (3)	$< .001$	**.258**
Neg. Attitude	AI Usage	6.95 (3)	.073	.080
Trust	Gender	0.33 (1)	.567	.004
Pos. Attitude	Gender	3.45 (1)	.063	.040
Neg. Attitude	Gender	0.15 (1)	.696	.002

- **Large Usage Effects ($\varepsilon^2 \approx .26$):** *Very Often* ($M_{\textbf{Trust}} = 4.42, M_{\textbf{Pos}} = 4.70$) and *Often* ($M_{\textbf{Trust}} = 4.07, M_{\textbf{Pos}} = 4.16$) > *Sometimes* ($M_{\textbf{Trust}} = 3.40, M_{\textbf{Pos}} = 3.64$).
- **Stable Caution:** Negative Attitude does *not* drop with heavier usage ($p = .073$).
- **Gender Null:** Consistent with López-Simó et al. (2026).

Figure 2
Mean Trust and Positive Attitude by AI Usage Frequency



Note. **DSCF post-hoc: Very Often & Often > Sometimes ($p \leq .014$).**

Bivariate Correlations: The Trust–Attitude Link

Table 7
Spearman's Rank-Order Correlations Between Study Constructs

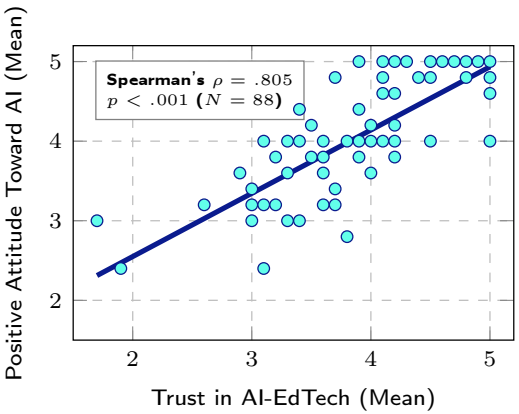
Construct Pair	Spearman's ρ	p
Trust ↔ Positive Attitude	**.805**	$< .001$
Trust ↔ Negative Attitude	.110	.309
Positive ↔ Negative Attitude	.028	.792

Note. $df = 86$ **for all pairs (**$N = 88$**).**

Theoretical Implications

- **Extended TAM Supported:** Strong positive association ($\rho = .805$) confirms trust and perceived usefulness travel together (Nazaretsky et al., 2025).
- **True Bidimensionality:** Near-zero correlation between Positive and Negative Attitude ($\rho = .028$) proves orthogonal dimensions (Schepman & Rodway, 2026).

Figure 3*Scatterplot of Trust Against Positive Attitude*



Note. **Each point represents one participant (**$N = 88$**).**

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Discussion: Trust as Prerequisite vs. The Danger of Over-Trust

Cross-Cultural Transferability

- Scales validated in UK general-public (Schepman & Rodway, 2026) and Swiss university (Nazaretsky et al., 2025) contexts transfer cleanly to **East Malaysian pre-university physics**.
- Addresses the regional validation gap flagged by Agyare et al. (2025) and Azizi et al. (2026).

Why Residual Caution Persists

- Heavy users show high trust (4.42) yet maintain moderate negative concerns (3.23).
- Aligns with **ambivalence theory** (Schneider & Schwarz, 2017) and European physics cohorts (López-Simó et al., 2026; Veith et al., 2026).

The Over-Trust Risk in Physics

- **Goal is Appropriate Reliance**, not maximal trust (Lee & See, 2004).
- Awareness of ethical/epistemic obstacles does *not* automatically dampen trust (Nazaretsky et al., 2025).
- Students offload cognitive work to ChatGPT even when aware of hallucination risks (Yeung et al., 2026; Lai et al., 2024).
- High baseline trust ($M = 3.92$) + novice prior knowledge = high vulnerability to **fluent, authoritative-sounding physics errors** (Dahlkemper et al., 2023).

Curriculum & Pedagogical Recommendations

Moving Beyond Generic "AI Literacy" to Physics-Specific Verification Training:

1. AI Problem-Auditing Tasks

Embed exercises where students receive an AI-generated physics solution and must actively locate embedded flaws (Dahlkemper et al., 2023; Veith et al., 2026):

- Dimensional inconsistencies
- Incorrect unit conversions
- Misapplication of Newton's laws
- Scalar–vector confusions

2. Structured Verification Heuristics

Require mandatory physical sanity checks before accepting any AI output in coursework (Lee & See, 2004; Marimon et al., 2025; Mustofa et al., 2025):

- **Limiting-case reasoning** (e.g., behaviour as $\theta \to 0$ or $m \to \infty$)
- **Order-of-magnitude estimation**
- Calibration of risk perception alongside self-efficacy (Zhang & Yu, 2025)

Target Outcome: Shift students from *affect-based over-reliance* to *calibrated, critical verification*.

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Methodological Limitations & Future Research

Study Limitations

- **Sample Size & Scope:** Modest sample ($N = 88$) from a single Sarawak Matriculation College; "Rarely" usage cell is very small ($n = 3$).
- **Exploratory Scope:** EFA only; requires independent replication for Confirmatory Factor Analysis (CFA).
- **Partial Trust Construct:** Administered Usefulness and Trust/Intent (10 items), omitting Moral Obstacles and Readiness (Nazaretsky et al., 2025).
- **Item Deviations:** Unchanged B6 wording and simplified C10 clause.

5 Directions for Future Research

1. **Pedagogical Pilots:** Test physics AI-auditing and verification tasks in coursework.
2. **Multi-Site CFA:** Replicate across Malaysian matriculation colleges with larger samples.
3. **Full 4D Trust SEM:** Administer all four dimensions (Nazaretsky et al., 2025) and test causal pathways via SEM.
4. **Module Comparisons:** Compare *Sains Fizikal* vs. *Sains Hayat* streams.
5. **Longitudinal Tracking:** Examine whether trust recalibrates or erodes on advanced physics problems (Wattanakasiwich et al., 2025).

Conclusion

Summary of Contributions

1. **Validated Instruments (RQ1):** Both the physics-adapted Short GAAIS-10 ($\alpha = .825$–$.902$) and AI-EdTech Trust Scale ($\alpha = .921$) demonstrated clean two-factor structures and strong reliability in an East Malaysian matriculation cohort.
2. **Considered Optimism Profile (RQ2 & RQ3):** Students exhibit high Positive Attitude ($M = 4.08$) and high Trust ($M = 3.92$) alongside moderate, usage-invariant Negative Attitude ($M = 3.23$)—validating bidimensional attitude measurement (Schepman & Rodway, 2026).
3. **Extended TAM & Pedagogical Call:** Trust and Positive Attitude are tightly coupled ($\rho = .805$), underscored by large usage-frequency effects ($\varepsilon^2 \approx .26$). High baseline trust in a verification-heavy discipline demands explicit physics accuracy-verification training (Dahlkemper et al., 2023; Lee & See, 2004).

Thank You! Questions & Discussion

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